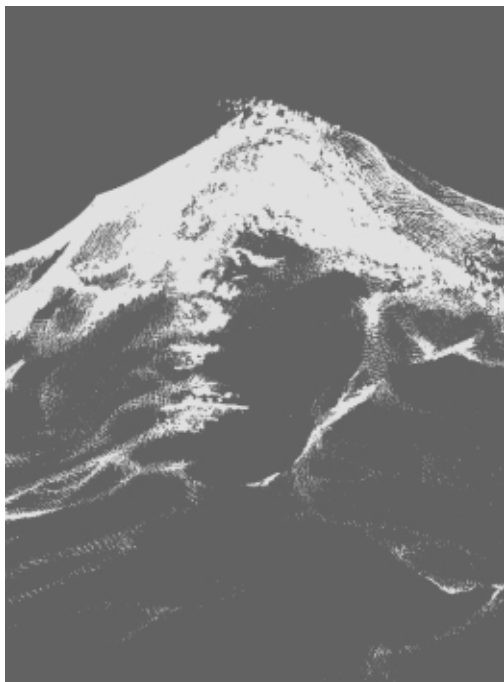


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## Den choice and occupation patterns of shelters by two sympatric lobster species, *Panulirus argus* and *Panulirus guttatus*, under experimental conditions

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**Abstract.** *Panulirus argus* and *P. guttatus* are sympatric and commonly share dens. We studied their den-related behaviour in 3 m-diameter tanks, each holding a casita (low den) and a cave (high den) (1) separately (2) together (3) together with a triggerfish predator (*Balistes vetula*), and (4) separately with the predator. Most *P. argus* left shelter at night, and some during the day, and showed no den preference in any treatment. In caves, individual *P. argus* occupied all parts of caves in treatments 1 and 2 and mostly the vault (walls and ceiling) in treatments 3 and 4. Not all *P. guttatus* left shelters at night, and none during the day. Individual *P. guttatus* showed no den preference in treatments 1 and 2, preferred the cave in 3 and 4, and always preferred the cave vault. Presence of congeners did not affect den choice or cave position for either species, whereas the predator affected den choice (*P. guttatus* only) and position in caves (both species). Videotapes showed aggression between individual *P. guttatus* but not between individual *P. argus* or between species. The fish cohabited with lobsters in the casita but attacked them in caves and in the open. Unlike *P. argus*, *P. guttatus* exhibited no communal defence.

### Introduction

Predation is one of the major forces controlling the distribution and abundance of prey, but prey animals can reduce the risk of predation by seeking appropriate refuge during the part of the day when predators are most active (Sih 1980, 1987; Hill and Wassenberg 1993; Tayasu *et al.* 1996). For example, many decapods including the spiny lobsters (Palinuridae) hide during the day in a variety of shelters and perform most of their activities at night (see, e.g., Phillips *et al.* 1980; Herrnkind *et al.* 1994), because most of their predators are diurnal fishes (Randall 1967; Heck and Weinstein 1989).

Sympatric species of the genus *Panulirus* may differ in habitat use. For example, in shallow areas of South Africa, *Panulirus gilchristii* inhabits rocky habitats, whereas *P. delagoae* occurs outside rocky areas (Berry 1971). Similarly, on the Mexican Pacific coast, *P. inflatus* occupies rocky bottoms whereas *P. gracilis* is mostly found on gravel-sand bottoms (Briones-Fourzán and Lozano-Álvarez 1992).

*Panulirus argus* and *P. guttatus* are sympatric throughout the Caribbean Sea (Holthuis 1991). The benthic phases of *Panulirus argus* undergo several ontogenetic shifts in habitat, from sea-grass and algal habitats to hard grounds with crevice-type shelters to reef tracts (Butler and Herrnkind 1997). In contrast, *P. guttatus* is restricted to the

reef habitat throughout its benthic life (Briones-Fourzán 1995; Sharp *et al.* 1997). Therefore, although *P. argus* and *P. guttatus* differ in habitat use, the two cohabit in coral reefs at some point in their lives, but their behavioural interactions as related to den resources are unknown.

The behaviour of *P. argus*, a highly migratory species, has been amply studied, including its diel rhythms (Kanciruk and Herrnkind 1973; Lipcius and Herrnkind 1982), den choice—either alone or under the risk of predation—(Eggleston *et al.* 1990, 1992; Eggleston and Lipcius 1992), social behaviour (Herrnkind 1980; Childress and Herrnkind 1996, 1997), and behaviour as related to predators (see, e.g., Herrnkind and McLean 1973; Kanciruk 1980; Lozano-Álvarez and Spanier 1997). In contrast, little is known about the diel rhythms and general behaviour of *P. guttatus*, other than that it is not migratory, is an obligate reef-dweller, exhibits group denning, and has well-developed homing and orientation capabilities (Caillouet *et al.* 1972; Herrnkind 1980; Carrasco-Zanini 1985; Negrete-Soto 1988; Briones-Fourzán 1995; Briones-Fourzán and McWilliam 1997; Sharp *et al.* 1997).

Although the two species share the coral-reef habitat, each apparently has certain specific den requirements. For example, *P. guttatus* is restricted either to shelters under coral heads deep enough for individuals to retreat into fully or to large crevices and caverns, where individuals are

usually found clinging to the ceiling or the walls. In contrast, *P. argus* can occupy shelters ranging from relatively small holes that conceal only the abdomen to large fissures and caverns, where individuals are mostly found on the floor (Sharp *et al.* 1997). At Looe Key, Florida, USA, Sharp *et al.* (1997) found that individual *P. argus* and *P. guttatus* rarely occupied a shelter at the same time, and when they did, individuals of each species tended to occupy the same position in the cave as when alone (i.e., *P. argus* the floor and *P. guttatus* the walls or ceiling). This result suggests that the occupation of a den by individuals of each species is independent of the presence of individuals of the other species and that their differences in den requirements may be one reason the two species can share den resources.

Communal sheltering is a general feature of palinurids (Atema and Cobb 1980; Kanciruk 1980; Zimmer-Faust and Spanier 1987), and it occurs in both *P. argus* and *P. guttatus*. This communal behaviour is exploited in several Caribbean lobster fisheries (Briones-Fourzán *et al.* 2000) by means of 'casitas', artificial shelters with a flat, low roof, but only *P. argus* is found beneath casitas (Briones-Fourzán 1995). Although casitas are typically deployed on vegetated areas, where *P. guttatus* does not dwell, in some areas they are deployed on hard grounds adjacent to coral reefs and yet fail to attract *P. guttatus* (Briones-Fourzán 1995). Under certain circumstances, therefore, the two species may compete for shelter, or *P. argus* may be able to exclude *P. guttatus* from low-roofed shelters such as casitas.

Because the two species are sympatric and in certain habitats—like patch reefs or small reef tracts—the number of dens may be limited, we experimentally explored den choice and shelter-occupation patterns by these species under several conditions (i.e., separately, together, together in the presence of a piscine predator, and separately in the presence of the predator) to find out whether individuals of these species share limited den resources, how they interact when the situation arises, and whether their specific den choice and occupation pattern change under the risk of predation.

## Materials and Methods

The experiments were conducted at the Puerto Morelos Academic Unit of the National University of Mexico, in Puerto Morelos, Quintana Roo, Mexico (20°51'N, 86°55'W). The experiments addressed the choice between a casita (i.e., a low, flat-roofed den) and a cave-like shelter (i.e. a high-ceilinged den) as well as the position of the lobsters sheltering in the cave, whether on the floor or clinging to the walls and ceiling of the cave. These issues were explored in four different treatments:

*Treatment 1:* each species tested separately (1a: eight *P. argus* per tank; 1b: eight *P. guttatus* per tank);

*Treatment 2:* the two species tested together (four lobsters of each species per tank).

*Treatment 3:* the two species tested together in the presence of a diurnal piscine predator, the triggerfish *Balistes vetula* (four lobsters of each species plus one triggerfish per tank);

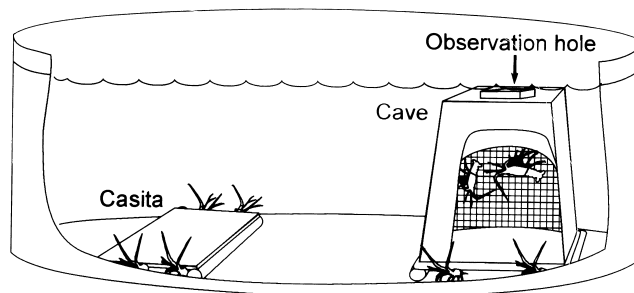
*Treatment 4:* each species tested separately in the presence of a triggerfish (4a: eight *P. argus* and a triggerfish per tank; 4b: eight *P. guttatus* and a triggerfish per tank).

Triggerfishes (Balistidae) are known to prey on juvenile and adult spiny and slipper lobsters, and have been used in predation experiments with several lobster species (Kanciruk 1980; Barshaw and Spanier 1994; Lozano-Álvarez and Spanier 1997).

During each treatment, we used four outdoor circular tanks 3 m in diameter and 1 m in depth. Each tank served as a replicate, so each treatment was replicated four times. A continuous seawater flow maintained ambient temperature in the tanks. The tanks were placed under a translucent shade and hence subject to ambient light. Two shelters were deployed inside each tank: (a) a flat-roofed casita, constructed of a ferrocement plate measuring  $0.8 \times 0.6$  m, bolted longitudinally to two PVC pipes 7.5 cm in diameter, and (b) a tall cave with a low entrance, constructed of an inverted plastic trash bin 0.82 m in height. The base of the cave was attached to PVC pipes like those used in the casita. The interior of the bin was lined with wire mesh slightly separated from the walls of the bin, to provide gripping surface for the lobsters (Fig. 1). The mesh reduced the inner height of the cave to about 0.50 m. The top of the cave had an 'observation hole', through which the number and position of lobsters in its interior could be observed. The hole was only uncovered while data were being recorded. The average illumination at 1200 hours (as measured with a diving PAM [pulse amplitude modulated] fluorometer calibrated against a LI-COR cosine-corrected light sensor), was  $240 \mu\text{mol m}^{-2} \text{s}^{-1}$  at the surface of the water in the tanks;  $63 \mu\text{mol m}^{-2} \text{s}^{-1}$  at the open bottom of the tanks;  $3.0 \mu\text{mol m}^{-2} \text{s}^{-1}$  at the centre point of the casita, and  $1.0 \mu\text{mol m}^{-2} \text{s}^{-1}$  at the centre point of the cave. Therefore, the illuminations within the casita and the cave were 4.7% and 1.6%, respectively, of the light penetrating to the bottom of the tank.

Lobsters were caught either with traps deployed in the coral reef at Puerto Morelos (*P. guttatus*) or by hand (*P. argus*). Some *P. argus*, and all the triggerfishes, were purchased live from local fishermen. All the specimens were acclimated for at least three days prior to each experiment; the two species were kept separately in acclimation tanks identical to the experimental tanks but without shelters.

A summary of relevant data on the lobsters used in each treatment (i.e., size range and sex ratio) is shown in Table 1. Although we planned to use lobsters of roughly the same size range in each treatment, we could not always do so. The size range of the triggerfish was 29.8–34.0 cm (fork length) in treatment 3, 28.0–36.0 cm in treatment 4a, and 31.0–37.0 cm in treatment 4b. Unfortunately, during treatment 3, the triggerfish in tank 4 jumped out of the tank between



**Fig. 1.** Schematic representation of the tanks used in the experiments on den choice and occupation pattern of caves.

**Table 1. Number of females (F) and males (M) and mean size (CL, carapace length in mm) of spiny lobsters, *Panulirus argus* and *P. guttatus*, in each of four tanks**

Treatments are: 1 = each species tested separately, (a) *P. argus*, (b) *P. guttatus*; 2 = the two species tested together; 3 = the two species tested together with one piscine predator, the triggerfish *Balistes vetula*; 4 = each species tested separately with one triggerfish, (a) *P. argus*, (b) *P. guttatus*

| Treatment 1               |     |                    | Treatment 2 |                    | Treatment 3 |                    | Treatment 4 |                    |
|---------------------------|-----|--------------------|-------------|--------------------|-------------|--------------------|-------------|--------------------|
| Tank                      | F:M | CL (mean $\pm$ SD) | F:M         | CL (mean $\pm$ SD) | F:M         | CL (mean $\pm$ SD) | F:M         | CL (mean $\pm$ SD) |
| <i>Panulirus argus</i>    |     |                    | (a)         |                    |             |                    |             |                    |
| 1                         | 4:4 | 87.4 $\pm$ 7.09    | 3:1         | 71.3 $\pm$ 5.04    | 3:1         | 59.3 $\pm$ 8.67    | 4:4         | 68.4 $\pm$ 4.50    |
| 2                         | 4:4 | 88.4 $\pm$ 8.47    | 3:1         | 70.7 $\pm$ 3.98    | 2:2         | 54.8 $\pm$ 1.45    | 2:6         | 69.0 $\pm$ 9.89    |
| 3                         | 4:4 | 92.5 $\pm$ 14.38   | 3:1         | 72.6 $\pm$ 1.72    | 1:3         | 62.1 $\pm$ 11.58   | 3:5         | 65.7 $\pm$ 10.60   |
| 4                         | 4:4 | 89.9 $\pm$ 13.84   | 2:2         | 77.5 $\pm$ 4.89    | 1:3         | 71.4 $\pm$ 10.89   | 4:4         | 69.5 $\pm$ 4.55    |
| <i>Panulirus guttatus</i> |     |                    | (b)         |                    |             |                    |             |                    |
| 1                         | 3:5 | 60.7 $\pm$ 5.98    | 2:2         | 56.6 $\pm$ 7.48    | 0:4         | 62.2 $\pm$ 7.51    | 4:4         | 52.7 $\pm$ 6.66    |
| 2                         | 2:6 | 46.6 $\pm$ 18.56   | 1:3         | 58.9 $\pm$ 3.51    | 1:3         | 66.6 $\pm$ 3.63    | 3:5         | 57.5 $\pm$ 9.35    |
| 3                         | 3:5 | 59.0 $\pm$ 13.50   | 1:3         | 59.2 $\pm$ 9.11    | 3:1         | 57.1 $\pm$ 6.63    | 20:6        | 54.8 $\pm$ 7.27    |
| 4                         | 3:5 | 56.0 $\pm$ 7.46    | 1:3         | 57.1 $\pm$ 11.84   | 2:2         | 55.3 $\pm$ 6.64    | 4:4         | 49.5 $\pm$ 10.38   |

**Table 2. Dates and relevant characteristics of observation periods for each treatment**

Treatments are: 1a = *Panulirus argus* alone, 1b = *P. guttatus* alone, 2 = the two species together, 3 = the two species together with one triggerfish predator, 4a = *P. argus* alone with a triggerfish, 4b = *P. guttatus* alone with a triggerfish

| Treatment | Period of observations | Range in sunrise time (h) | Range in sunset time (h) | Temperature in tanks (°C) at noon |
|-----------|------------------------|---------------------------|--------------------------|-----------------------------------|
| 1a        | 13–16 July 1999        | 06:14–06:15               | 19:33–19:32              | 27.7–29.1                         |
| 1b        | 8–11 Sept. 1999        | 06:33–06:34               | 18:57–18:55              | 30.1–31.0                         |
| 2         | 16–20 Nov. 1999        | 06:58–07:02               | 18:06–18:05              | 25.3–26.5                         |
| 3         | 12–16 April 2000       | 06:31–06:28               | 19:05–19:07              | 26.7–28.0                         |
| 4a        | 10–14 Oct. 2000        | 06:41–06:43               | 18:27–18:24              | 26.1–27.0                         |
| 4b        | 30 Sep.–04 Oct. 2000   | 06:39–06:40               | 18:36–18:32              | 28.0                              |

observation periods on day 1 and died, so data from only three tanks were analysed in treatment 3.

Before the beginning of each experiment, half of the lobsters needed for each experimental tank were randomly chosen from the acclimation tank and placed inside a mesh cylinder 0.5 m in diameter located equidistant from the two shelters in the experimental tank. The other half was placed in a similar mesh cylinder opposite to the first cylinder. At 1800 hours, the cylinders confining the lobsters were lifted and removed from the tank. In treatments 3 and 4, one fish was introduced in a third mesh cylinder at the centre of each tank. The fish was released one hour later than the lobsters. The observations began the following morning. Lobsters were fed daily with mussels and clams, which were introduced into each tank at 1830 hours on days 2–4. The fish were kept partially starved; they were fed one or two unshelled mussels per day. Because during treatment 3 the fish ate the food intended for the lobsters, we did not provide any lobster food during treatment 4. At the end of each experiment, the lobsters were tagged and released on a coral reef, 4 km from the lab, to ensure that they would not be used again. New lobsters and fish were used in each treatment.

Observations were conducted every four hours, starting at 0600 hours, for four consecutive days. Table 2 shows the dates of each treatment, as well as the range in sunrise and sunset times and the

maximum temperature in the tanks measured at noon in each treatment. To determine the sheltering schedule (i.e., the time spent within either shelter) of each lobster species in the different treatments, we recorded the number of lobsters outside both shelters and inside the cave in each tank at every observation hour. In the case of *P. argus*, individuals beneath the casitas were easily counted because of their exposed antennae or cephalothoraces, but the *P. guttatus* were difficult to count because they retreated fully under the casitas. Therefore, the *P. guttatus* that were neither in the cave nor in the open were assumed to be beneath the casita. The count was verified at the end of each treatment series. The position of the lobsters inside the cave (floor, walls or ceiling) was also recorded. Initially, we intended to analyse separately the number of lobsters on the walls and on the ceiling, but in treatments 2 and 3 the number of lobsters of each species in the tanks was relatively small and the large number of zero values in either of the two positions precluded statistical analyses, so for these purposes we combined the wall and ceiling data and reduced the positions inside the cave to two: floor and 'vault'. When lobsters were observed in the open, the type of activity they were performing, i.e., walking, resting, or exhibiting communal defensive displays such as groups, rosettes, or queues (as described by Herrnkind 1980), was recorded as well.

On the basis of results of the sheltering schedule, the daily data obtained at 1000 hours—when the majority of lobsters were denning—

were used to analyse the den choice and the occupation pattern of the caves by the lobsters. We analysed den choice in each of the four treatments for each lobster species by testing the heterogeneity of the data due to observation day (1–4) and to replicate tank (1–4) with independent G-tests (Sokal and Rohlf 1981). The average distribution of lobsters in all tanks and on all days represented a conservative estimate of the den occupation pattern and was tested against a random pattern of den occupation, also with a G-test. Then, to determine whether the pattern of den occupation by each lobster species changed in response to the presence of the congener and/or the triggerfish predator, we conducted a comparison across the four treatment combinations with G-tests. This procedure was repeated for the occupation patterns of caves exhibited by each lobster species.

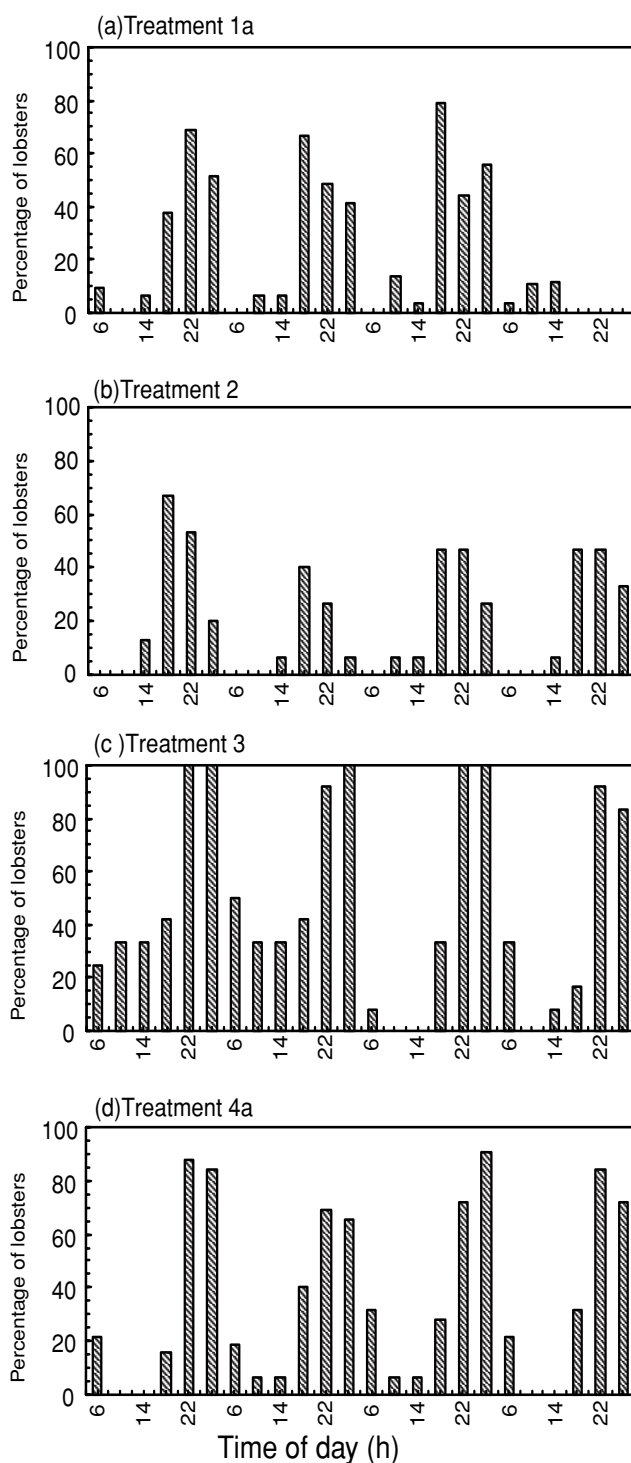
A video camera was fixed to the observation hole at the top of each cave during treatment 3, to obtain information on the aggressive inter- and intraspecific interactions among lobsters and fish within the caves. A second hole held a lamp that produced a dim red light with a wavelength  $>600$  nm. Although the activity inside the caves was continuously recorded for 80 hours, we only analysed 10 min of recordings per hour between 0600 and 1900 hours, when the majority of lobsters were sheltered. We estimated the percentage of time the lobsters engaged in aggressive interactions, such as grabbing, pushing, fending and chasing (as described by Atema and Cobb 1980). Aggressive interactions between individual *P. guttatus*, individual *P. argus*, and individuals of different species were separated. Also, the percentage of time occupied by fish attacks and lobster defensive displays inside the caves was analysed.

## Results

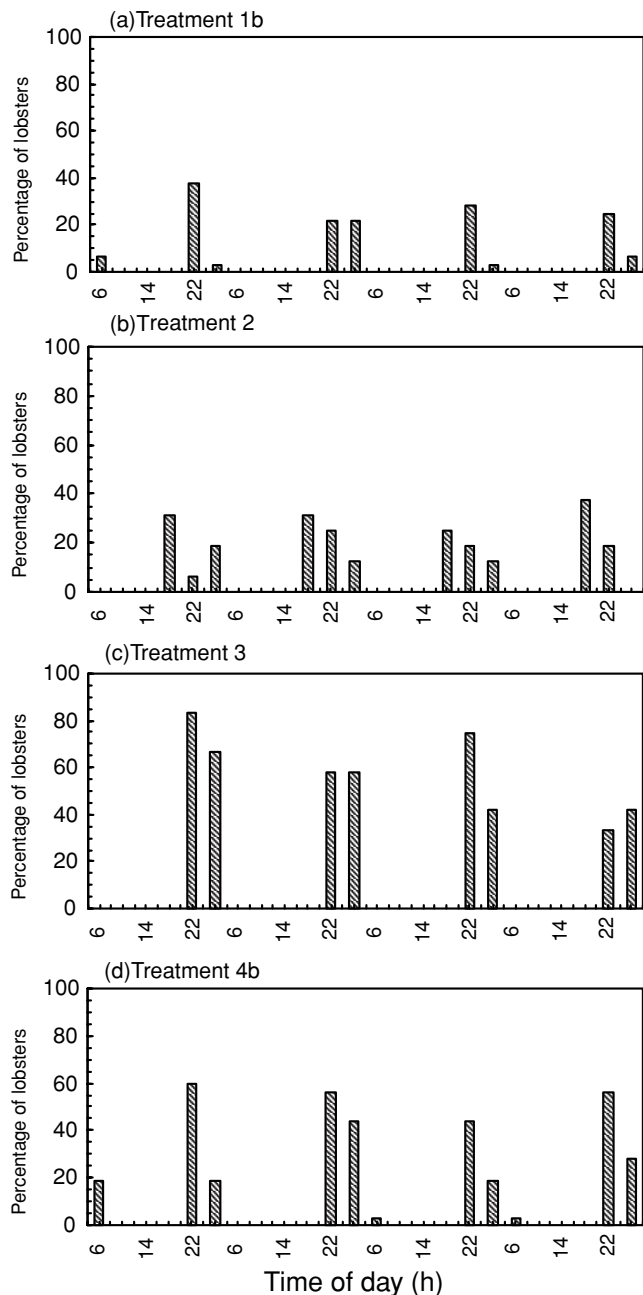
### Sheltering schedule

During treatments 1a and 2, a large percentage of individual *P. argus* were outside the shelters from shortly before sunset through the night, but a few individuals were also observed outside the shelters during the day (Fig. 2a, 2b). However, *P. argus* increased the time spent outside the shelters during treatments 3 and 4a (in the presence of a predator), when most individuals (up to 100% in treatment 3) were observed in the open at night, and a large percentage of lobsters was also observed outside the shelters throughout the day (Fig. 2c, 2d). The fish were observed attacking groups of two or more lobsters in the open during the day. The unsheltered lobsters were typically forming groups or walking around the tank in queues during the daytime, whether the fish attacked them or not, and these types of communal behaviour were also common at night.

*Panulirus guttatus* spent most of the time in the shelters during treatment 1; less than 40% of individuals were outside the shelters in the dark hours of the night (Fig. 3a). This sheltering schedule of *P. guttatus* remained similar during treatments 2, 3, and 4b (Fig. 3b, 3c, 3d), but the percentage of individuals observed outside the shelters at night was higher in those treatments that included a triggerfish (Fig. 3c, 3d), especially when the triggerfish was inside the cave. Individual *P. guttatus* were never observed in the open during the day, and unsheltered individuals at night were dispersed throughout the tanks without any evidence of group formation nor displays of communal behaviour.



**Fig. 2.** Sheltering schedule of *Panulirus argus* during each of the four treatments. The bars represent the percentage of lobsters outside the shelters in each observation period (every 4 h, from 0600 hours on day 1 to 0200 h on day 4, except in treatment 1a, when the experiment was terminated at 1400 hours on day 4). Composite data for four tanks, except in treatment 3 (three tanks). Treatment conditions are 1a = alone ( $n = 32$ ), 2 = with congener ( $n = 16$ ), 3 = with congener and triggerfish predator ( $n = 12$ ), and 4a = alone with triggerfish ( $n = 32$ ).



**Fig. 3.** Sheltering schedule of *Panulirus guttatus* during each of the four treatments. The bars represent the percentage of lobsters outside the shelters in each observation period (every 4 h, from 0600 hours on day 1 to 0200 hours on day 4). Composite data of four tanks, except in treatment 3 (three tanks). Treatment conditions are 1b = alone ( $n = 32$ ), 2 = with congener ( $n = 16$ ), 3 = with congener and triggerfish predator ( $n = 12$ ), and 4b = alone with triggerfish ( $n = 32$ ).

### Den choice

*Panulirus argus* showed no significant preference for either the casita or the cave in any of the four treatments (Table 3). Day heterogeneity in treatments 1a and 4a reflects the movement by individuals from one type of shelter to the

other between days. Tank heterogeneity in most treatments indicates that in some tanks individuals favoured one of the two types of shelter. The average distribution indicates that individual *P. argus* occupied the two shelter types randomly regardless of the presence of *P. guttatus* and/or the triggerfish predator. Individual *P. argus* sheltering beneath the casitas typically had their antennae or their whole cephalothoraces exposed.

*Panulirus guttatus* favoured the cave over the casita in all treatments (Table 3), but this den choice was only significant in treatments 3 and 4b, when a predator was present. Individual *P. guttatus* sheltering beneath the casitas were completely hidden by the flat roof and were not visible from above.

When we compared the average den-choice pattern of *P. argus* and *P. guttatus* across the four treatment combinations, it further became clear that *P. argus* remained neutral in shelter-type preference in the presence and absence of *P. guttatus* and/or the triggerfish predator (Table 4). The den choice of *P. guttatus* changed significantly in the presence of a predator, and the den choice of each species was significantly different under the risk of predation. The rest of the comparisons (Table 4) showed no significant change in den choice.

### Occupation patterns of the cave

*Panulirus argus* showed no significant preference for either the floor or the vault of the cave in treatments 1a and 2 (Table 5), although in treatment 1a twice as many *P. argus* were on the floor as were on the vault. The day heterogeneity in treatment 1a shows that individuals moved between the two places in time. In treatment 3, no *P. argus* were observed on the floor, and in treatment 4a significantly more individuals occupied the vault of the cave, except in one tank. In contrast, *P. guttatus* showed a marked preference for the vault of the cave in all treatments, particularly in treatments 3 and 4b, in which no *P. guttatus* were observed on the floor (Table 5).

The crossed comparisons among all treatments (Table 6) show that the occupation pattern of the cave by *P. argus*—either alone or with *P. guttatus*—changed significantly when the predator was around and that *P. argus* and *P. guttatus* exhibited different occupation patterns when each of these species was held separately in the absence of a predator. The remaining comparisons did not evidence a significant effect on the occupation patterns of caves by either species.

### Inter- and intraspecific aggressive interactions

In total, 500 minutes of videotape recorded during treatment 3 were analysed. Lobsters sharing the caves showed relatively few aggressive interactions during the observation periods (Fig. 4). All the aggressive interactions occurred between individual *P. guttatus*, and none was recorded

**Table 3. Den choice (casitas vs. caves) of individual *P. argus* and *P. guttatus* at 1000 hours on each of four days in four replicate tanks**

Average distribution of lobsters in all tanks and days represents a conservative estimate of the den occupation pattern and was tested against a random pattern of den occupation by G-test. The hypothesis of heterogeneity was examined by both observation day and tank replicate by G-test. Treatment conditions for each species are: 1 = alone ( $n = 32$ ), 2 = with congener ( $n = 16$ ), 3 = with congener and triggerfish predator ( $n = 12$ ), and 4 = alone with triggerfish predator ( $n = 32$ )

| Species            | Treatment | Comparison           | G-score | P       | Casitas | Caves |
|--------------------|-----------|----------------------|---------|---------|---------|-------|
| <i>P. argus</i>    | 1a        | Day heterogeneity    | 9.095   | < 0.05  |         |       |
|                    |           | Tank heterogeneity   | 13.875  | < 0.005 |         |       |
|                    |           | Average distribution | 0.143   | n.s.    | 13      | 15    |
| <i>P. argus</i>    | 2         | Day heterogeneity    | 1.311   | n.s.    |         |       |
|                    |           | Tank heterogeneity   | 3.475   | n.s.    |         |       |
|                    |           | Average distribution | 0.604   | n.s.    | 6       | 9     |
| <i>P. argus</i>    | 3         | Day heterogeneity    | 0.178   | n.s.    |         |       |
|                    |           | Tank heterogeneity   | 14.984  | < 0.001 |         |       |
|                    |           | Average distribution | 0.403   | n.s.    | 4       | 6     |
| <i>P. argus</i>    | 4a        | Day heterogeneity    | 14.098  | < 0.005 |         |       |
|                    |           | Tank heterogeneity   | 7.860   | < 0.025 |         |       |
|                    |           | Average distribution | 0.291   | n.s.    | 14      | 17    |
| <i>P. guttatus</i> | 1b        | Day heterogeneity    | 8.658   | < 0.05  |         |       |
|                    |           | Tank heterogeneity   | 90.297  | < 0.001 |         |       |
|                    |           | Average distribution | 2.021   | n.s.    | 12      | 20    |
| <i>P. guttatus</i> | 2         | Day heterogeneity    | 2.051   | n.s.    |         |       |
|                    |           | Tank heterogeneity   | 30.581  | < 0.001 |         |       |
|                    |           | Average distribution | 2.306   | n.s.    | 5       | 11    |
| <i>P. guttatus</i> | 3         | Day heterogeneity    | 1.247   | n.s.    |         |       |
|                    |           | Tank heterogeneity   | 32.454  | < 0.001 |         |       |
|                    |           | Average distribution | 5.822   | < 0.025 | 2       | 10    |
| <i>P. guttatus</i> | 4b        | Day heterogeneity    | 4.639   | n.s.    |         |       |
|                    |           | Tank heterogeneity   | 3.313   | n.s.    |         |       |
|                    |           | Average distribution | 29.399  | < 0.001 | 2       | 30    |

**Table 4. Average den choice pattern of *P. argus* and *P. guttatus* compared across four treatment combinations by G-test**  
Treatment conditions are 1 = alone ( $n = 32$ ), 2 = with congener ( $n = 16$ ), 3 = with congener and triggerfish predator ( $n = 12$ ), and 4 = alone with triggerfish predator ( $n = 32$ )

| Species                                          | Social condition                 | Predator condition           | G      | P       |
|--------------------------------------------------|----------------------------------|------------------------------|--------|---------|
| <i>P. argus</i>                                  | alone (1a) vs. with congener (2) | absent                       | 0.164  | n.s.    |
| <i>P. argus</i>                                  | alone (4a) vs. with congener (3) | present                      | 0.082  | n.s.    |
| <i>P. argus</i>                                  | alone                            | absent (1a) vs. present (4a) | 0.010  | n.s.    |
| <i>P. argus</i>                                  | with congener                    | absent (2) vs. present (3)   | 0.000  | n.s.    |
| <i>P. guttatus</i>                               | alone (1b) vs. with congener (2) | absent                       | 0.184  | n.s.    |
| <i>P. guttatus</i>                               | alone (4b) vs. with congener (3) | present                      | 1.032  | n.s.    |
| <i>P. guttatus</i>                               | alone                            | absent (1b) vs. present (4b) | 9.938  | < 0.005 |
| <i>P. guttatus</i>                               | with congener                    | absent (2) vs. present (3)   | 0.803  | n.s.    |
| <i>P. argus</i> (1a) vs. <i>P. guttatus</i> (1b) | alone                            | absent                       | 0.490  | n.s.    |
| <i>P. argus</i> (2) vs. <i>P. guttatus</i> (2)   | with congener                    | absent                       | 0.259  | n.s.    |
| <i>P. argus</i> (4a) vs. <i>P. guttatus</i> (4b) | alone                            | present                      | 13.751 | < 0.001 |
| <i>P. argus</i> (3) vs. <i>P. guttatus</i> (3)   | with congener                    | present                      | 1.508  | n.s.    |

**Table 5. Occupation patterns of caves (floor or vault, i.e. walls + ceiling) of individual *P. argus* and *P. guttatus* at 1000 hours on each of four days in four replicate tanks**

Average distribution of lobsters in all tanks and days represents a conservative estimate of the cave occupation pattern and was tested against a random pattern of cave occupation by G-test. The hypothesis of heterogeneity was examined by both observation day and tank replicate by G-test. In those cases where the G-test was not possible as a result of zero values, the results were graded as 'similar', when the data had the same tendency, or 'different', when the data did not have the same tendency. Treatment conditions for each species are: 1 = alone ( $n = 32$ ), 2 = with congener ( $n = 16$ ), 3 = with congener and triggerfish predator ( $n = 12$ ), and 4 = alone with triggerfish predator ( $n = 32$ )

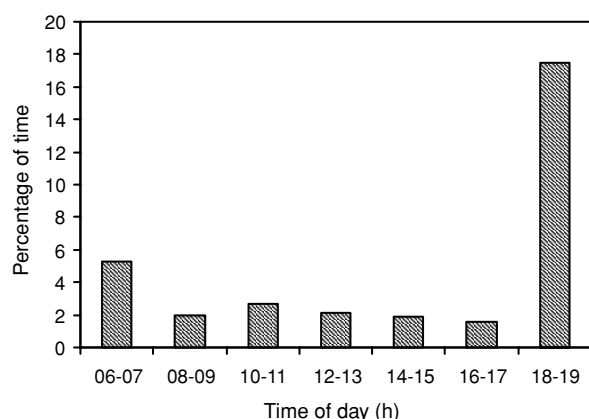
| Species            | Treatment | Comparison           | G-score | <i>P</i>  | Floor | Vault |
|--------------------|-----------|----------------------|---------|-----------|-------|-------|
| <i>P. argus</i>    | 1a        | Day heterogeneity    | 22.263  | < 0.001   |       |       |
|                    |           | Tank heterogeneity   | 1.560   | n.s.      |       |       |
|                    |           | Average distribution | 1.699   | n.s.      | 10    | 5     |
| <i>P. argus</i>    | 2         | Day heterogeneity    | 1.669   | n.s.      |       |       |
|                    |           | Tank heterogeneity   | 1.648   | n.s.      |       |       |
|                    |           | Average distribution | 0.111   | n.s.      | 4     | 5     |
| <i>P. argus</i>    | 3         | Day heterogeneity    | –       | similar   |       |       |
|                    |           | Tank heterogeneity   | –       | similar   |       |       |
|                    |           | Average distribution | –       | 0.000     | 0     | 6     |
| <i>P. argus</i>    | 4a        | Day heterogeneity    | 1.965   | n.s.      |       |       |
|                    |           | Tank heterogeneity   | –       | different |       |       |
|                    |           | Average distribution | 7.723   | <0.01     | 3     | 14    |
| <i>P. guttatus</i> | 1b        | Day heterogeneity    | 2.954   | n.s.      |       |       |
|                    |           | Tank heterogeneity   | 1.770   | n.s.      |       |       |
|                    |           | Average distribution | 14.723  | <0.001    | 2     | 18    |
| <i>P. guttatus</i> | 2         | Day heterogeneity    | 1.412   | n.s.      |       |       |
|                    |           | Tank heterogeneity   | 1.961   | n.s.      |       |       |
|                    |           | Average distribution | 4.818   | <0.05     | 2     | 9     |
| <i>P. guttatus</i> | 3         | Day heterogeneity    | –       | similar   |       |       |
|                    |           | Tank heterogeneity   | –       | similar   |       |       |
|                    |           | Average distribution | –       | 0.000     | 0     | 10    |
| <i>P. guttatus</i> | 4b        | Day heterogeneity    | –       | similar   |       |       |
|                    |           | Tank heterogeneity   | –       | similar   |       |       |
|                    |           | Average distribution | –       | 0.000     | 0     | 30    |

**Table 6. Average occupation pattern of caves for *P. argus* and *P. guttatus* compared accross four treatment combinations by G-test**

In cases where the G-test was not possible because of zero values, the results were graded as 'similar', when the data of the compared treatments had the same tendency, and 'different', when they did not. Treatment conditions for each species are: 1 = alone ( $n = 32$ ), 2 = with congener ( $n = 16$ ), 3 = with congener and triggerfish ( $n = 12$ ), and 4 = alone with triggerfish predator ( $n = 32$ )

| Species                                          | Social condition                 | Predator condition           | G      | <i>P</i>  |
|--------------------------------------------------|----------------------------------|------------------------------|--------|-----------|
| <i>P. argus</i>                                  | alone (1a) vs. with congener (2) | absent                       | 1.141  | n.s.      |
| <i>P. argus</i>                                  | alone (4a) vs. with congener (3) | present                      | –      | similar   |
| <i>P. argus</i>                                  | alone                            | absent (1a) vs. present (4a) | 8.290  | < 0.005   |
| <i>P. argus</i>                                  | with congener                    | absent (2) vs. present (3)   | –      | different |
| <i>P. guttatus</i>                               | alone (1b) vs. with congener (2) | absent                       | 0.408  | n.s.      |
| <i>P. guttatus</i>                               | alone (4b) vs. with congener (3) | present                      | –      | similar   |
| <i>P. guttatus</i>                               | alone                            | absent (1b) vs. present (4b) | –      | similar   |
| <i>P. guttatus</i>                               | with congener                    | absent (2) vs. present (3)   | –      | similar   |
| <i>P. argus</i> (1a) vs. <i>P. guttatus</i> (1b) | alone                            | absent                       | 12.905 | < 0.001   |
| <i>P. argus</i> (2) vs. <i>P. guttatus</i> (2)   | with congener                    | absent                       | 1.639  | n.s.      |
| <i>P. argus</i> (4a) vs. <i>P. guttatus</i> (4b) | alone                            | present                      | –      | similar   |
| <i>P. argus</i> (3) vs. <i>P. guttatus</i> (3)   | with congener                    | present                      | –      | similar   |





**Fig. 4.** Percentage of time occupied by lobsters in aggressive interactions within the caves during treatment 3 (*Panulirus argus*, *P. guttatus*, and the triggerfish predator). Composite data for three tanks and four days of video recordings (total time: 500 min). The interior of the caves was recorded for 10 min every hour between 0600 and 1900 hours, and the data were grouped in two-hour periods (i.e., each 2-h period on the X-axis represents 20 min of recordings). Only aggressive interactions between individuals of *P. guttatus* occurred.

between individual *P. argus* or between individuals of different species. The interactions between *P. guttatus* occurred mostly around the time of sunset (17.9% of the recorded time in the period from 1800 to 1900 hours), when the general activity in the cave increased, or around sunrise, when most of the lobsters returned to the shelters, although a few also occurred at other times of day (Fig. 4).

Most aggressive interactions involved individual *P. guttatus* near the ceiling of the cave. The interactions were typically started by a large male. For example, we observed several events in which a large male climbed to the ceiling and began pushing and chasing other smaller *P. guttatus*, which responded by descending to the walls or by fleeing all the way outside the cave. The aggressive individuals of *P. guttatus* never attacked directly any individuals of *P. argus* even when the latter were also on the ceiling. During several aggressive encounters, individuals of *P. guttatus* walked over individuals of *P. argus*, but the latter did not respond and did not appear disturbed by the activity of the *P. guttatus*. In general, however, individual *P. argus* mostly occupied the walls, whereas individual *P. guttatus* tended to seek the ceiling, i.e., the highest portion of the cave vault.

Eleven fish attacks were recorded inside the caves. These attacks amounted to only 8.25 (1.6%) of the 500 minutes of video recordings. Typically, the fish swam into the cave and up to the ceiling in an almost vertical, head-up position, directing its attack at the antennae and the eyes of the lobsters clinging to the ceiling, which were mostly *P. guttatus*. The lobsters engaged in defensive behaviour such as antenna sweeping and fending, successfully keeping

the fish at bay. Only those individuals directly attacked responded. The rest of the lobsters retreated as far as possible from the fish and remained still. No lobster died in any tank as a consequence of a fish attack during the experiment.

## Discussion

Although most lobsters are active nocturnally (Phillips *et al.* 1980), not all palinurid species are exclusively nocturnal. For example, *P. argus* commonly performs some sort of diurnal activity, both under experimental conditions (Lipcius and Herrnkind 1982; Glaholt 1990; Lozano-Álvarez 1996; Lozano-Álvarez and Spanier 1997) and in the field. Moreover, *P. argus* undergoes diurnal massive migrations (Herrnkind 1980) and mates in the open in broad daylight (Lipcius *et al.* 1983; Lipcius and Herrnkind 1985; Lozano-Álvarez *et al.* unpubl. data). In our tanks, most individual *P. argus* sheltered during the day when alone or when cohabiting with *P. guttatus*, but this activity pattern of *P. argus* was seemingly disrupted by the presence of the diurnal predator. Instead of denning to avoid the predator, many individual *P. argus* remained in the open in broad daylight and commonly exhibited communal defensive displays such as the forming of pods, groups, and queues, whether the triggerfish was attacking them or not. Although differences in photoperiod and water temperature could partially explain these changes in activity patterns of *P. argus* (Kanciruk and Herrnkind 1973; Lipcius and Herrnkind 1982), these abiotic features were similar during treatments 1a and 3 (see Table 2).

On average, *P. argus* did not show any preference between the caves and the casitas in any treatment, but in some tanks most individuals occupied the casita whereas in other tanks they occupied the cave. Apparently then, sheltering was more important to *P. argus* than the physical attributes of the shelter. Moreover, individual *P. argus* occupying the casitas, and those occupying the floor of the caves, were commonly observed with their antennae or their whole cephalothoraces exposed. These results corroborate the findings of Herrnkind *et al.* (1975) and Sharp *et al.* (1997) that *P. argus* has a wide den preference and can use any crevice-type shelter as long as it is able to conceal the abdomen, and they support the hypothesis of Childress and Herrnkind (1997) that den-seeking individuals prefer to enter a den already occupied by a conspecific, which presumably reduces their exposure in the open to predators. This intraspecific attraction appears to be chemically mediated through odours released by conspecifics (Ratchford and Eggleston 1998, 2000).

In the caves, the average occupation pattern by *P. argus* was random when alone or when cohabiting with *P. guttatus*, but in the presence of the triggerfish predator, most individuals shifted to the walls or ceiling of the cave, suggesting an avoidance of the predator. Although the

differential occupation of the walls and ceiling of the cave by individual lobsters was not statistically analysed, we did observe that, when alone with the predator, individual *P. argus* consistently tended to occupy the ceiling of the cave. In contrast, in the video recordings during treatment 3, most individual *P. argus* were on the walls and most *P. guttatus* were clinging to the ceiling, suggesting some kind of interference competition for space between *P. argus* and *P. guttatus* in the presence of the triggerfish.

When a triggerfish approached a casita, it was repelled by the fast movement and sweeping of the antennae of the *P. argus* hiding beneath, but eventually the fish was able to enter the shelter, where it cohabited with the lobsters without attacking them. Lozano-Álvarez and Spanier (1997) hypothesized that, even when *P. argus* and its predators (triggerfishes, nurse sharks, etc.) commonly cohabit beneath casitas in the field (Eggleston and Lipcius 1992; Arce *et al.* 1997; Eggleston *et al.* 1997; Cruz and Phillips 2000), casitas protect lobsters against diurnal fish predators because the fish need open space to attack the lobsters. In contrast, when a fish entered a cave, it occasionally attacked the lobsters clinging to the ceiling (whether *P. argus* or *P. guttatus*), which kept the fish at bay by vigorously sweeping or raking their antennae. Even though our caves had a larger inner space than the casitas, this space was apparently still small enough to prevent the fish from killing any lobster. In contrast, in coral-reef habitats dens may vary in size from small holes and narrow fissures to large and deep caverns, and *P. argus* is mostly found on the floor of the caves (Sharp *et al.* 1997). Large caves may allow for a communal defence by *P. argus*, which displays this behaviour when individuals are on the floor (Herrnkind *et al.* 1975; Lozano-Álvarez and Spanier 1997). The displacement of individual *P. argus* to the walls and ceilings of our caves in the presence of the triggerfish was probably due to the low entrance of the caves relative to their height, which enabled the fish to enter and attack the lobsters from within. In this situation, remaining on the floor would not allow individual *P. argus* to defend themselves against the fish.

In the field, individual *P. guttatus* are rarely observed outside their shelters during the day but are commonly seen crawling on coral reefs and boulder jetties at night (Caillouet *et al.* 1972; Sharp *et al.* 1997; Lozano-Álvarez *et al.* unpubl. data). In our tanks, *P. guttatus* was certainly more reclusive than *P. argus*. Individual *P. guttatus* abandoned their shelters only during the dark part of the night, and none was ever recorded in the open in daylight whether alone or with *P. argus*. Even at night, many individuals remained in the shelters, at least during the observation periods. When the triggerfish were also present, the activity pattern of *P. guttatus* did not change, but the proportion of individuals of *P. guttatus* that left the shelters at night was higher. On some occasions, the fish were found inside the cave at night, apparently asleep, but in contrast to

*P. argus*, unsheltered *P. guttatus* did not show any tendency to group or form queues. On the contrary, individual *P. guttatus* were scattered throughout the tank.

*Panulirus guttatus* showed, on average, a greater affinity for the caves than for casitas in the four treatments, although this affinity was not significant in treatments 1b and 2. This result suggests that the minimal difference in light intensity between shelter types on the one hand and the sole cohabitation with *P. argus* on the other, did not necessarily influence the den choice of *P. guttatus*. However, the presence of the predator significantly increased the preference of *P. guttatus* for the caves. Moreover, *P. guttatus* rarely occupied the floor of the caves in any treatment, but in the presence of *P. argus* and the predator (treatment 3), individual *P. guttatus* tended to shift to the highest part of the vault (ceiling).

Tayasu *et al.* (1996) found that the risk of predation reduced the intra- and interspecific interference competition for refuge in two species of hippolytid shrimps by lowering the activity of both species, but Richards and Cobb (1986) concluded that competition for dens between Jonah crabs (*Cancer borealis*) and lobsters (*Homarus americanus*)—especially when habitat is limited—would affect lobsters more because the crabs have a wider habitat use. Similarly, competition for dens between *P. argus* and *P. guttatus* would be expected to affect *P. guttatus* more seriously than *P. argus*, but in our treatment 3 (i.e. in a den-limited situation and under the risk of predation) the differential occupation of the vault of the cave (more *P. guttatus* on the ceiling and more *P. argus* on the walls) suggests that competition for dens between the two species is somewhat buffered by their differential occupation of the caves. Aggressive interspecific interactions were not ever recorded, however, so the possibility of an interspecific chemical-mediated interaction between *P. argus* and *P. guttatus* should be explored.

The domination of small by larger individuals of the same species has been well documented (e.g., by Wilson 1975; Werner and Gilliam 1984). During treatment 3, individual *P. guttatus* in the cave fought for the occupation of the ceiling, and the largest males clearly dominated the smaller individuals. These interactions between individual *P. guttatus* in caves also occur in the absence of both *P. argus* and the triggerfish (Lozano-Álvarez and Segura-García unpubl. data). In the field, Sharp *et al.* (1997) observed large males of *P. guttatus* seemingly guarding dens to protect harems of females from other males. Although this behaviour has also been observed in *P. argus* (Lipcius *et al.* 1983; Lipcius and Herrnkind 1985), we did not record any size-related dominance in *P. argus* in our caves (but this result could be due to the relatively small sizes of the *P. argus* in treatment 3; see Table 1), and individual *P. argus* over a wide size range have been observed cohabiting beneath casitas in the field (Briones-

Fourzán *et al.* 2000). *Panulirus guttatus* may therefore have a more hierarchical social organization than *P. argus* in relation to shelter occupancy.

In summary, the presence of a predator had more effect on the activity pattern, den choice, and occupation pattern of *P. argus* and *P. guttatus* than did cohabitation solely with the other lobster species, but the two species showed different specific behavioural traits. For example, *P. argus* (a) was more active and more flexible in den use than *P. guttatus*, in both the presence and the absence of a predator; (b) increased its diurnal activity in the presence of a predator; and (c) exhibited communal defensive behaviour both in the shelters and in the open. In contrast, *P. guttatus* (a) was less active and more reclusive than *P. argus*; (b) only left the shelters in the dark part of the night; (c) significantly preferred the caves over the casitas in the presence of a predator; (d) had a greater affinity for the vault than for the floor of the caves in all treatments; (e) showed no communal defensive displays whether in the open or in the shelters; and (f) seemed to establish an intraspecific size-related hierarchy in communal dens. Therefore, *P. argus* relies more on communal defence against predators in shelters such as casitas in the form of group exposure and use of antennae in whipping movements, and on the formation of groups, pods, rosettes, and queues to fend off attacks from predators in the open, whereas the defensive strategy of *P. guttatus* involves remaining in shelters in daylight and preferentially in those shelters that allow individuals to conceal their whole bodies and to retreat as far back as possible to deter the attack of a predator. In the presence of a predator, each species tends to occupy a different part of the den but apparently without interspecific aggressive interactions. In the field, where the risk of predation is permanent, these differences between *P. argus* and *P. guttatus* may partially explain (a) why *P. guttatus* does not occupy casitas adjacent to coral reefs and (b) how the two species coexist in the coral reef habitat and share habitat resources even when dens are limited.

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